

Debounce vs Throttling in JavaScript

1. Definitions

Debounce: Ensures a function is only executed after a specified delay period has passed since the last call. Good for scenarios where you want to wait until the user has finished an action.

Throttling: Ensures a function is executed at most once in a specified time interval, no matter how many times it's called. Good for scenarios where continuous execution needs to be controlled.

2. Common Use Cases

- Debouncing:
 - Search box input (API calls after user stops typing)
 - Window resize (avoid firing resize handler on every pixel change)
 - Auto-save feature triggered after typing stops
- Throttling:
 - Scroll event handling (e.g., infinite scroll, animation triggers)
 - Mouse move events (drag & drop, drawing on canvas)
 - Button click spam prevention (limit rapid clicks)

3. Code Examples

```
// Debounce Implementation
const debounce = (func, delay) => {
  let timer;
  return function(...args) {
    clearTimeout(timer);
    timer = setTimeout(() => {
      func.apply(this, args);
    }, delay);
  };
};

// Usage
const handleSearch = debounce((query) => {
  console.log("API Call with query:", query);
}, 1000);

// Throttle Implementation (timestamp-based)
const throttle = (func, delay) => {
  let lastCall = 0;
  return function(...args) {
    const now = Date.now();
    if (now - lastCall >= delay) {
      lastCall = now;
      func.apply(this, args);
    }
  };
};

// Usage
const logScroll = throttle(() => {
  console.log("Scroll event at:", Date.now());
}, 2000);
```

4. Key Differences

- Debouncing waits until the user stops triggering the event.
- Throttling executes immediately and then ensures execution at most once in a given interval.
- Debounce is delay-focused (collects burst into one execution).
- Throttle is interval-focused (regular execution with gaps).

5. Common Interview Questions

1. What is the difference between debouncing and throttling?
2. Give real-world use cases for debouncing and throttling.
3. How would you implement debounce and throttle in JavaScript?
4. When would you choose debounce over throttle?
5. What's the difference between timestamp-based and timeout-based throttling?